Introduction to Computer Science

18th November 2024

Examination Paper

Answer ALL questions.

Clearly cross out any surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination.

SECTION A - Multiple Choice Question

Answer these in your answer booklet – do NOT write on this Question Paper.
Write ONE (1) correct answer from A, B, C, or D for each question in your answer booklet.
Each question is worth 1 mark.

Question 1
What is an embedded system?

A A computer system not linked to another computer system

B A device that contains a computer, such as a car or washing machine

C A computer system that undertakes most frequent tasks required by general users

D Two or more computer systems linked together

1 mark

Question 2
What is the purpose of a fan on a CPU?

A To speed up the CPU's processing power

B To increase the computer's overall storage capacity

C To reduce the noise made by the CPU

D To prevent the CPU from overheating

1 mark

Question 3
What does a computer’s clock regulate and how is its speed measured?

A It regulates memory storage and is measured in bits per second

B It controls CPU cooling and is measured in watts

C It regulates the timing and speed of all functions, measured in Megahertz (MHz) or Gigahertz (GHz)

D It manages data transmission between devices and is measured in gigabytes

1 mark

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Marks

Question 4
What is Network Attached Storage (NAS)?

A A device directly connected to a computer to store data

B A network hard drive that can store, retrieve and share data over a network

C A type of cloud storage used only for businesses

D A storage system that cannot be accessed from home

1 mark

Question 5
What is software piracy?

A Legally purchasing software and sharing it with friends

B Unauthorised use, reproduction or distribution of software for profit

C Using free software provided by the developer

D Downloading software from a licensed vendor

1 mark

Question 6
What is an operating system that allows more than one program to run simultaneously called?

A Multiprocessing operating system

B Multithreading operating system

C Real-time operating system

D Multitasking operating system

1 mark

Question 7
What is the result of an XOR operation if both input bits are the same?

A 0

B 1

C The sum of the input bits

D The product of the input bits

1 mark

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Marks

Question 8
How does a WAP function within a network?

A It directly connects devices to the internet.

B It acts as a firewall for all connected devices.

C It creates a wireless network and forwards traffic to the main wired network.

D It stores data from wireless devices

1 mark

Question 9
What is the difference between download and upload speeds in broadband internet?

A Download speed measures how fast you can receive data, and upload speed measures how fast you can send data.

B Download speed measures how fast you can send data, and upload speed measures how fast you receive data.

C Upload speeds are always faster than download speeds.

D Download speeds are measured in Mb/s. Upload speeds are measured in Kb/s

1 mark

Question 10
How does a SQL injection attack compromise a website or application?

A By manipulating the query to gain unauthorised access to the database

B By crashing the server hosting the application

C By stealing data directly from the web server’s file system

D By altering the website's appearance without database interaction

1 mark

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Marks

Question 11
What is ‘big data’?

A Data that requires to be saved on optical storage

B An increase in the volume of data stored and analysed by computer systems

C A type of cloud storage used for small data sets

D A method of processing data used in network protocols

1 mark

Question 12
How does increasing a computer's RAM improve its performance?

A By speeding up the internet connection

B By improving the graphics rendering speed

C By enhancing the computer’s ability to run multiple tasks simultaneously

D By increasing the storage capacity for files

1 mark

Question 13
What is a HDMI cable primarily used for?

A Transmitting internet data between devices

B Connecting devices for data storage

C Transmitting digital audio and video between devices

D Powering electronic devices

1 mark

Question 14
Which register holds the memory address of the next instruction to be fetched from primary memory?

A Memory Address Register (MAR)

B Program Counter (PC)

C Memory Data Register (MDR)

D Accumulator (ACC)

1 mark

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Marks

Question 15
What is the purpose of defragmentation software?

A It deletes old files from the hard drive to free up space.

B It compresses files to save storage space.

C It backs up files to prevent data loss.

D It rearranges file segments so they are stored contiguously, improving performance.

1 mark

Question 16
What is the decimal equivalent of the binary number 11100000?

A 128

B 192

C 224

D 256

1 mark

Question 17
What is a key characteristic of bitmap (raster) images?

A They are made up of pixels arranged in horizontal rows

B They use mathematical equations to represent shapes.

C They cannot be edited unless converted to a PNG first

D They are unsuitable for photographs and image retouching.

1 mark

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Marks

Question 18
Which of the following best describes bit depth in digital sound?

A The number of bits used per second, measured in kilobits per second (kbps).

B The number of sound samples taken per second.

C The number of bits available for each sample, with higher bit depth resulting in better sound quality.

D The number of audio channels in a recording.

1 mark

Question 19
What is the value of output Q if the three inputs (R,S,T) in the logic statement below all have the initial value of (ONE) 1?

Q = NOT(R) AND (S OR T)

A 0

B 1

C 2

D -1

1 mark

Question 20
Which of the following is an example of cloud storage?

A Google Drive

B Local Hard Drive

C USB Flash Drive

D DVD

1 mark

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Marks

Question 21
Which of the following correctly demonstrates the input, process and output in a supermarket checkout system?

A Input: The cashier counts the items manually.
Process: The computer calculates the total.
Output: The receipt is printed.

B Input: The computer finds the total.
Process: The cashier scans the items.
Output: The total is displayed on the screen.

C Input: Staff use a scanner to input the item.
Process: The computer finds the price from a database and adds it to a total.
Output: The total is shown on the screen.

D Input: The customer pays for the items.
Process: The cashier checks for discounts.
Output: The computer updates its inventory.

1 mark

Question 22
How is a bus organised in a computer system?

A It only transmits addresses between components.

B It uses control lines for sending data and data lines for commands only.

C It has control lines for signalling and data lines for transmitting data and addresses.

D It does not include a process for sending or receiving data.

1 mark

Question 23
Which type of wearable device is specifically designed to be worn on the wrist and offers features like fitness tracking and app notifications?

A VR Headset

B Hearables

C Smart Clothing

D Smartwatches

1 mark

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Marks

Question 24
What is a processor, also known as a central processing unit (CPU)?

A The main memory of the computer

B The main chip in a computer system located on a motherboard

C A device used for cooling the computer

D A storage device for saving data

1 mark

Question 25
What is the effect of having more cores in a CPU?

A It doubles the number of instructions that can be processed.

B It allows the CPU to process more instructions, but not necessarily double.

C It reduces the processing power of the CPU.

D It decreases the number of fetch-decode-execute (FDE) cycles.

1 mark

Question 26
What is the smallest unit of measurement in computing?

A Nibble

B Kilobyte

C Byte

D Bit

1 mark

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Marks

Question 27
Which of the following is an example of a second-generation programming language?

A Binary machine code

B Python

C JavaScript

D Assembly Language

1 mark

Question 28
What is the purpose of Plug and Play in modern operating systems?

A It allows users to manually configure new hardware.

B It enables automatic detection and configuration of new hardware peripherals.

C It disables hardware detection to increase system speed.

D It automatically updates all software installed on the computer.

1 mark

Question 29
What is the process for representing a negative integer in two's complement notation?

A Write the number in binary, invert the bits and add 1

B Invert the number and subtract 1

C Write the number in binary and add 1 to the leftmost bit

D Subtract the binary number from 11111111

1 mark

Question 30
Why is Unicode more suitable for representing characters from different languages compared to older character encoding systems?

A It uses 8 bits and supports 256 characters.

B It only supports English and a few European languages.

C It supports over 65,000 characters, making it suitable for multiple languages.

D It requires less memory than ASCII and includes emojis.

1 mark

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Marks

SECTION B
ANSWER ALL QUESTIONS

Question 31

a) A hospital vending machine provides hot and cold drinks.

Customers select their choice of drink from a menu on screen and provide payment through a chip and pin card reader.

When payment has been taken the machine dispenses the drink and displays a message on the screen thanking the customer.

i) Describe the vending machine system in terms of input, process and output. **3**

ii) State the type of interface the vending machine uses to interact with customers (shown below). **1**



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iii) State the name of TWO (2) INPUT devices used in the system. **2**

iv) Last year the hospital employed a person to serve the drinks. **4**

Describe ONE (1) advantage and ONE (1) disadvantage of replacing the human worker with the drinks vending machine.

Total 10 Marks

Question 32

a) A police station records all suspect interviews as audio files with maximum file size 100MB.

i) Calculate the maximum number of audio files they could store on a 1TB external hard drive. You must show your working. **2**

ii) Explain the term 'solid-state' memory. **2**

iii) State ONE (1) disadvantage of using solid state storage compared to magnetic tape. **1**

iv) The police have decided to try Cloud storage. **4**

Describe ONE (1) advantage **and** ONE (1) disadvantage of using Cloud storage.

v) The audio of each interview is converted into a lossy audio file. **1**

State ONE (1) possible file format for the recordings.

Total 10 Marks

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Question 33

a) Explain what is meant by the term ‘software’?+ **1**

b) For each example of software given below state whether it is application software or utility software.

i) CAD **1**

ii) E-Mail **1**

iii) Web Browser **1**

iv) Anti-virus **1**

c) Describe how an Operating System supports the following aspects of computer usage:

i) Interrupt Handling **2**

ii) Managing Memory **2**

d) State ONE (1) feature of an Operating System that uses a WIMP Interface. **1**

Total 10 Marks

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Marks

Question 34

a) Complete the table by converting the 8-bit binary number into decimal and the decimal into 8-bit binary. **3**

8-bit Binary	Decimal
11110000	
	106
00001111	

b) Complete the table by writing the answer for each statement. **3**

Statement		Answer
(i)	The binary code that represents the hexadecimal value ‘B’	
(ii)	The smallest hexadecimal value that can be represented using 5-bits	
(iii)	The maximum number of different colours that can be represented in a PNG with a colour depth of 7 bits	

c) Perform a TWO (2) place binary shift to the left on the value 00011010 and describe the mathematical effect of this operation. **2**

d) Explain why binary addition may lead to an overflow error. **2**

Provide an example to support your answer.

Total 10 Marks

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Marks

Question 35

a) A new office has installed a network which includes FOUR (4) computers and a switch. There are TWO (2) printers that can be accessed by all the computers.

The computers and printers are arranged in a star topology.

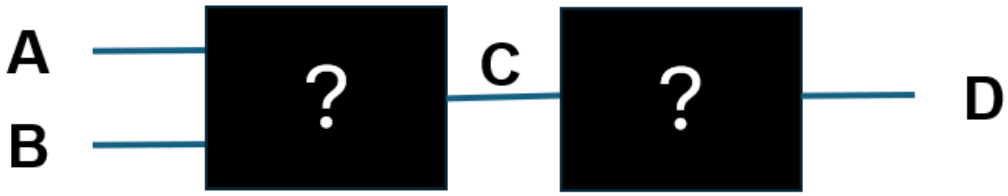
i) Draw a diagram to show how the computers, printers and switch are connected in a star topology. **3**

ii) Describe the role of the switch in a star network. **2**

iii) State ONE (1) advantage of using a star topology over a ring topology. **1**

b) A logic diagram is given below without the logic gates shown.

The truth table for the diagram is also provided.



Input A	Input B	C	D
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

i) Redraw the logic diagram using the correct symbols in place of the two unknown boxes. **3**

ii) Describe in words the input conditions that would turn on a system that uses this logic sequence. **1**

Total 10 Marks

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Question 36

a) State TWO (2) ethical issues in Computing. **2**

b) Describe TWO (2) ways the growth of computer usage has had a **positive** impact on the environment. **4**

c) Employers are often concerned with the health of their employees if they are sat using a computer for long periods of time.

i) Describe ONE (1) health problem associated with working on computers for a long period of time. **2**

ii) State TWO (2) ways in which users can prevent health problems when using a computer. **2**

Total 10 Marks

Question 37

a) Describe how the World Wide Web works. **8**

You must include the following terminology in your answer:

- URL
- DNS
- IP address
- Web Server
- HTML

b) Explain the term ‘phishing’ in relation to network security. **2**

Total 10 Marks

End of paper

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